

Public awareness of the need for security in computing systems is growing as critical services are becoming increasingly dependent on interconnected computing systems. National infrastructure components such as the electric power, telecommunication and transportation systems can no longer function without networks of computers. The advent of the World Wide Web has especially increased public concern for security. Security is the primary concern of businesses that want to use the Internet for commerce and maintaining business relationships.

The increased awareness of the need for security has resulted in an increase of efforts to add security to computing environments. However, these efforts suffer from the flawed assumption that security can adequately be provided in application space without certain security features in the operating system. In reality, operating system security mechanisms play a critical role in supporting security at higher levels. Yet today, debate persists in the research community as to what role operating systems should play in secure systems. The computer industry has not accepted the critical role of the operating system to security, as evidenced by the inadequacies of the basic protection mechanisms provided by current mainstream operating systems.

The necessity of operating system security to overall system security is undeniable; the underlying operating system is responsible for protecting application-space mechanisms against tampering, bypassing, and spoofing attacks. If it fails to meet this responsibility, system-wide vulnerabilities will result.

The need for secure operating systems is especially crucial in today's computing environment. Substantial increases in connectivity and data sharing have increased the risk to systems such that even a careful and knowledgeable user running on a single-user system is no longer safe from the threat of malicious code. Because the distinction between data and code is vanishing, malicious code may be introduced, without a conscious decision on the part of a user to install executable code, whenever data is imported into the system. For example, malicious code